

Kaveen Sheron Jayasinghe

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Education

- B.Sc. Engineering (Hons) in Computer Science & Engineering, University of Moratuwa — 3rd Year Undergraduate (2023 – Present).
- General Certificate Exam Advanced Level, physical sciences stream, Sri Lanka. ZScore +2.2905, with 3As (2022).
- General Certificate Exam Ordinary Level, 9As (2019).
- Primary and Secondary education at St. Joseph's College Colombo-10.

Experience

Collaborator, Oppia Foundation (2026 - Present)

- Member of the CORE (Creators, Operations, Reviewers, and Editors) team at the Google backed open source organization Oppia Foundation.
- Contribute to full stack feature development and resolve regressions across the project.
- Beyond feature development, act as a training co-lead by mentoring and onboarding new contributors, reviewing proposed solutions, assigning issues, and collaborating with project leads to coordinate quarterly planning and track project progress.

Associate Software Engineer, ictfromabc (2025 - Present)

- Developed and shipped features for a Laravel, SvelteKit, and Flutter based learning platform serving 55,000+ registered users.
- Implemented key platform features including password recovery, payment integration, ticketing system, user reviews with admin approval workflow and an SMS portal.
- Improved application quality by writing controller and integration tests for multiple backend modules and contributing to CI/testing workflows.

Sales Executive, Abans PLC (2022 -2023)

- Completed technical training on Apple devices and retail operational workflows.
- Achieved over LKR 20 million in sales in a single month by advising customers across multiple product categories.
- Strengthened communication and problem-solving skills in a high-volume retail environment.

Projects

Prypiatos

[link to the project](#)

- Led the design and deployment of a smart energy platform integrating ESP32 based edge nodes, MQTT telemetry pipelines, and cloud native backend services.
- Built MQTT based edge device workflows for real time energy data acquisition, local buffering, reconnection handling and reliable device-to-cloud message delivery.
- Engineered infrastructure and deployment pipelines using Terraform, Ansible, Docker, Kubernetes and Argo CD for automated provisioning, cluster setup, and application rollout.
- Defined Git based development workflows and drove cross team integration, release coordination and platform stabilization across the deployment lifecycle.

Cross-Lingual Speech & Identity Synthesis System

[link to the project](#)

- End-to-end English-to-Sinhala AI system for speech processing and NLP with speaker-preserving voice and lip synchronization.
- Built using WhisperX for transcription, NLLB-200 for translation, Coqui TTS and a custom voice conversion model to mimic the original speaker's voice, and Wav2Lip for realistic lip-syncing.
- Fully implemented in Python as an automated multimedia processing pipeline.

Autonomous Drone Platform for IESL-RoboGames

[link to the project](#)

- Contributed to a containerized sandbox environment for autonomous drone development using Docker, supporting Pixhawk, ArduPilot, and Raspberry Pi based workflows.
- Engineered isolated multi-user development containers with SSH access, enabling teams to work independently within dedicated user environments on shared infrastructure.
- Designed persistent storage and recovery mechanisms so user data, code, and workspace state remained intact across container restarts or failures, improving reliability of the development setup.

Veri5

[link to the project](#)

- Architected a full-stack platform using Next.js 16, React 19, Express 5, TypeScript, Prisma, and PostgreSQL, with modular APIs for authentication, test-kit verification, appointments, orders, clinics, and practitioner workflows.
- Implemented a client-side validation pipeline with TensorFlow.js and custom EXIF metadata parsing to run image inference in-browser, detect screenshots or edited uploads, and preserve privacy by keeping raw images off the backend.
- Designed transactional backend workflows with JWT and Google OAuth, hashed one-time status-share tokens, serial-based test-kit verification, and atomic database updates for kit usage, verification state, and appointment slot availability; selected as a Finalist at Devthon 3.0.

Ohana Note

[link to the project](#)

- Designed and implemented a multi service browser game application for secure message sharing, using UUID-keyed records to generate unique recipient-specific game links.
- Built the system using KAPLAY, Node.js, Express, MySQL, and Docker Compose, separating the landing page, backend API, game client, and database into independently deployed services.

N1K0

[link to the project](#)

- Advanced micromouse robot powered by ESP32 microcontroller and a custom-designed PCB.
- Features real-time sensor integration and autonomous maze navigation.

Kandypack

[link to the project](#)

- Supply chain management web application built with React, Express, and MySQL.

Open Source Contributions

- Actively contributing to the Oppia Foundation.
- Contributed to WSO2 open source projects during Hacktoberfest 2025.

Leadership & Volunteering

Selection Committee Lead — SLIoT 2026

- Headed participant selection and evaluation for a National level IoT and innovation-focused program, ensuring fair and technical assessment standards.

Program Committee Lead — Hit the Grounds 2025

- Led the planning and execution of an industry-facing competition with participation from 20 leading companies.

Technical Committee Member — IESL Robogames 2025/26

- Contributed to team evaluations, competition environment setup, and sandbox development as part of the Technical Committee for IESL Robogames 2025/26.

Member — AIESEC in Colombo South (2025–2026)

- Contributed to cross-cultural experience programs and internal operations within a global student-led organization.

Content Creator — Debug Diaries (2025)

- Handled graphic design, live filming, and real-time video editing for technical content.

Secretary — Science Union, St. Joseph's College, Colombo (2022)

- Managed meeting records, coordination, and administrative communication for the school science society.

Interests & Achievements

- Finalist: Devthon 3.0 (2026)
- Finalist: CodeRally (2025)
- Finalist: CyberShield 4.0 (2025)
- Contributor: Asia-Pacific Robot Contest (2025); Winner: Hitachi High-Tech Award
- Sports: Volleyball (All-Island Champion 2019), Cricket (2012–2022)

Technical Skills

Programming: C, C++, JavaScript, Python, SQL, Bash

Web/Frameworks: REST API, React, Express, Laravel, SvelteKit

Cloud & DevOps: Git, Docker, Kubernetes, ArgoCD, Terraform, Ansible, CI/CD, AWS (S3, EC2)

Other Skills

Languages: English (Professional working proficiency), Sinhala (Native proficiency)